Investigation of Effective State Representation for Side-Scan Sonar SLAM Under Elevation Degeneracy

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Abstract : *Side-scan sonar (SSS) has emerged as a promising sensing modality for underwater simultaneous localization and mapping (SLAM) because of its long-range sensing capability and wide-area seabed coverage. In landmark-based SSS SLAM, salient seabed features are typically represented using slant-range and azimuth observations derived from acoustic measurements. However, due to the sensing geometry of SSS, landmark elevation remains only weakly constrained by the measurements. This limitation, referred to in this work as elevation degeneracy in SSS SLAM, can increase estimation errors when conventional full 6-DOF formulations are directly adopted. This study investigates how landmark elevation uncertainty influences state estimation in landmark-based SSS SLAM under elevation degeneracy. In particular, Jacobian analysis of the slant-range and azimuth observation model is performed to examine how native SSS measurements contribute to each component of the vehicle state and influence the overall estimation behavior. Based on this analysis, an effective vehicle state representation is identified according to the state sensitivities imposed by the sensing geometry. The effectiveness of the resulting reduced 4-DOF state representation, which excludes roll and pitch, is validated through comparative physics-based underwater simulation experiments against a conventional 6-DOF formulation.*

1. Introduction

Side-scan sonar (SSS) has been widely utilized in underwater robotics due to its capability to provide long-range acoustic measurements with wide-area seabed coverage. Unlike conventional imaging sensors, SSS fundamentally acquires one-dimensional (1D) acoustic backscatter signals, referred to as pings, in which re flected intensity is measured primarily as a function of slant range [1, 2]. In many existing applications, these sequential ping measurements are stacked over time to construct two-dimensional (2D) acoustic images, en abling image-domain interpretation of underwater environments. Based on such image representations, visual landmarks or salient acoustic features can be extracted and incorporated into simultaneous localization and mapping (SLAM) frameworks for underwater navigation and mapping.

Several existing studies have explored SSS-based SLAM using image-level feature extraction approaches. Fallon et al. [3] fused acoustic ranging and SSS mea surements within a graph-based SLAM framework, while Zhang et al. [4] utilized visual feature correspondences extracted from stacked SSS imagery for pose-graph optimization. These approaches typically rely on ground range projection and altitude-dependent geometric approximations to associate image-domain features with three-dimensional (3D) landmark representations. Although effective in many practical scenarios, such for mulations inherently involve ambiguity in the vertical direction because conventional SSS measurements do not directly provide elevation information.

In feature-poor or homogeneous seabed environments, where reliable image-domain feature extraction becomes difficult, directly utilizing native slant-range measure ments at the ping level can provide a more suitable al ternative to conventional image-level formulations [5]. Compared to stacked-image-based approaches, ping-level processing preserves the original sensing geometry of SSS measurements and avoids additional distor tions introduced during image formation or ground-range projection. However, directly incorporating slant-range based landmark observations into a SLAM framework also exposes a fundamental estimation issue that becomes more explicit at the ping-measurement level. In particular, uncertainty in landmark vertical positions can propagate into the vehicle state estimation process, particularly under full 6-degrees-of-freedom (DOF) formulations. This issue, referred to in this work as elevation degeneracy in SSS SLAM (Fig.1), can induce unstable coupling between the vehicle's vertical position (typically, depth from water surface) and pitch states, ultimately increasing estimation errors.

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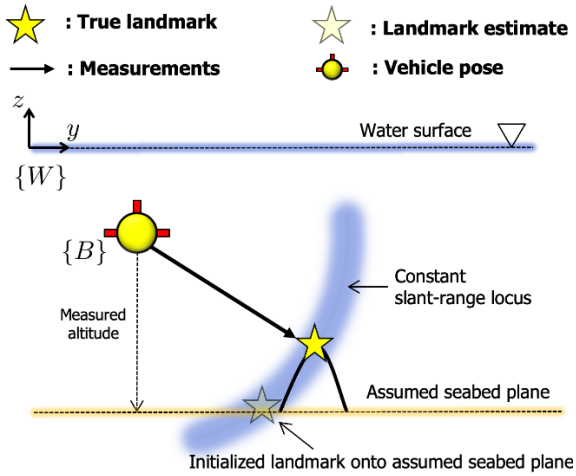


Fig. 1 Illustrative example of elevation degeneracy in SSS SLAM shown in the frontal view of the global frame $\{W\}$. From a single vehicle pose, an SSS measurement constrains the landmark to lie along a constant slant-range locus, leaving its elevation insufficiently constrained. In many SSS SLAM frameworks, the initialized landmark position may deviate from the true landmark location due to the inherent elevation ambiguity of the measurement.

Despite its practical importance, the effect of elevation degeneracy on state estimation in SSS SLAM has received limited explicit attention in the literature. While some approaches adopt reduced-dimensional state representations [6], the relationship between the SSS measurement model and the resulting estimation behavior has not been systematically analyzed. In this paper, we examine how elevation uncertainty in ping-level SSS SLAM affects the state sensitivity and estimation accuracy. In particular, we discuss the limitations of full 6-DOF state representations and consider a reduced 4-DOF state representation comprising horizontal position, depth, and yaw. The effect of elevation degeneracy under different state formulations is further evaluated through comparative physics-based simulation studies.

2. Problem Formulation

This study considers a landmark-based SSS SLAM framework in which seabed features observed by SSS are represented as landmarks in the environment map. By directly utilizing slant-range and azimuth measurements with in the native sensing geometry of SSS, the elevation-related degeneracy inherent in SSS observations

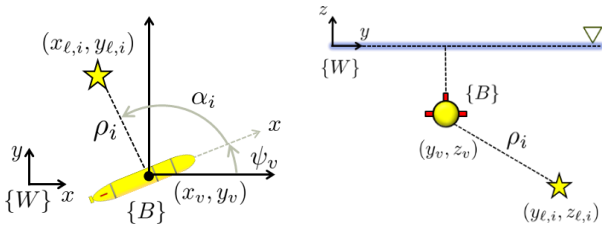


Fig. 2 Coordinate frames and measurement geometry of SSS observations under the reduced 4-DOF vehicle state representation. The vehicle pose (x_v, y_v, z_v, ψ_v) and the landmark position $(x_{\ell,i}, y_{\ell,i}, z_{\ell,i})$ are represented in the global frame. The measurements ρ_i and α_i denote the slant range and azimuth, respectively. The azimuth measurement is primarily associated with the horizontal relative geometry between the vehicle and the landmark, while the slant-range measurement reflects their full 3D separation.

can be more explicitly analyzed. The system state consists of the vehicle state and the landmark states. The full state vector at time step k defined as

$$\mathbf{x}_k = [\mathbf{x}_v \quad \mathbf{x}_{\ell,1} \quad \mathbf{x}_{\ell,2} \quad \cdots \quad \mathbf{x}_{\ell,i} \quad \cdots \quad \mathbf{x}_{\ell,n_\ell}]^\top, \tag{1}$$

where $\mathbf{x}_v$ denotes the vehicle pose state, and $x_{\ell,i} = [x_{\ell,i}, y_{\ell,i}, z_{\ell,i}]^\top$ represents the 3D position of the ith landmark. The stochastic measurement model using the slant-range and azimuth observations obtained from the ith landmark is expressed as

$$\mathbf{z}_i = \begin{bmatrix}\rho_i \\ \alpha_i\end{bmatrix} = \begin{bmatrix}h_1(\mathbf{x}_v, \mathbf{x}_{\ell,i}) \\ h_2(\mathbf{x}_v, \mathbf{x}_{\ell,i})\end{bmatrix} + \mathbf{v} = \begin{bmatrix}\sqrt{\Delta x_i^2 + \Delta y_i^2 + \Delta z_i^2} \\ \text{atan2}(\Delta y_i, \Delta x_i)\end{bmatrix} + \mathbf{v} \tag{2}$$

where $\boldsymbol{v} \sim \mathcal{N}(0, \mathrm{R})$ refers to the zero-mean Gaussian measurement noise vector, and $(\Delta x_i, \Delta y_i, \Delta z_i)$ denotes the relative position of the i th landmark in the vehicle body frame. Let $(x_v y_v, z_v)$ denote the vehicle position in the global frame. The relative landmark position in the global frame is then defined as

$$\begin{bmatrix}\delta x_i \\ \delta y_i \\ \delta z_i\end{bmatrix} = \begin{bmatrix}x_{\ell,i} - x_v \\ y_{\ell,i} - y_v \\ z_{\ell,i} - z_v\end{bmatrix}. \tag{3}$$

This relative position depends on the vehicle orientation represented by roll ϕ_v, pitch θ_v, and yaw ψ_v. The corresponding body-frame relative position is obtained as

$$\begin{bmatrix}\Delta x_i \\ \Delta y_i \\ \Delta z_i\end{bmatrix} = \mathcal{R}(\phi_v, \theta_v, \psi_v) \begin{bmatrix}\delta x_i \\ \delta y_i \\ \delta z_i\end{bmatrix} \tag{4}$$

where $\mathcal{R}(\phi_v, \theta_v, \psi_v)$ denotes the rotation matrix parameterized by the vehicle orientation angles.

A complete representation of an unmanned underwater vehicle (UUV) pose in 3D space requires a 6-DOF state vector that includes both 3-axis position and orientation. In this study, however, we focus on examining how SSS slant-range and azimuth measurements influence the update of both the vehicle and landmark states in a landmark-based SLAM formulation. For this purpose, we first consider a full state vector $\mathbf{x}_k = [\mathbf{x}_v^\top \ \mathbf{x}_{\ell,i}^\top]^\top$ that includes a 6-DOF vehicle's pose $\mathbf{x}_v = [x_v \ y_v \ z_v \ \phi_v \ \theta_v \ \psi_v]^\top$ and a single landmark state $\mathbf{x}_{\ell,i} = [x_{\ell,i} \ y_{\ell,i} \ z_{\ell,i}]^\top$. Under the side-looking geometry of SSS, where $\Delta x_i \approx 0$, and under typical seabed-mapping conditions where the vehicle maintains small roll and pitch angles, i.e., $\phi_v \approx 0$ and $\theta_v \approx 0$, the measurement Jacobian can be approximated as

$$\mathrm{H}_i = \frac{\partial \mathbf{h}(\mathbf{x}_v, \ \mathbf{x}_{\ell,i})}{\partial \mathbf{x}_k} = \begin{bmatrix}\frac{\partial h_1(\mathbf{x}_v, \ \mathbf{x}_{\ell,i})}{\partial \mathbf{x}_k}^\top & \frac{\partial h_2(\mathbf{x}_v, \ \mathbf{x}_{\ell,i})}{\partial \mathbf{x}_k}^\top\end{bmatrix}$$
$$\approx \begin{bmatrix}\frac{-\delta x_i}{\gamma_i} & \frac{-\delta y_i}{\gamma_i} & \frac{-\delta z_i}{\gamma_i} & 0 & 0 & 0 & \frac{\delta x_i}{\gamma_i} & \frac{\delta y_i}{\gamma_i} & \frac{\delta z_i}{\gamma_i} \\ \frac{\delta y_i}{\Delta y_i^2} & \frac{-\delta x_i}{\Delta y_i^2} & 0 & 0 & \frac{\delta z_i}{\Delta y_i} & -1 & \frac{-\delta y_i}{\Delta y_i^2} & \frac{\delta x_i}{\Delta y_i^2} & 0\end{bmatrix} \tag{5}$$

where

$$\gamma_i = \sqrt{\delta x_i^2 + \delta y_i^2 + \delta z_i^2}\,. \tag{6}$$

The Jacobian H_i characterizes the sensitivity of the slant-range and azimuth measurements to the vehicle and landmark states. The zero entries corresponding to roll indicate that the SSS measurements provide no direct update for the roll state. In contrast, the pitch-related term in the azimuth Jacobian is proportional to the landmark's vertical relative position. This implies that pitch estimation is coupled with the estimated landmark elevation.

In conventional 3D landmark-based SLAM with range-bearing observations, the coupling between vehicle orientation and landmark position can provide meaningful geometric constraints. However, in SSS SLAM, landmark elevation is only weakly constrained by the sensing geometry of SSS measurements. Consequently, fluctuations in the estimated landmark elevation can propagate into the pitch-state update, potentially increasing estimation errors in a full 6-DOF formulation. To alleviate this issue, this study adopts a reduced 4-DOF vehicle pose representation consisting of 3D position and yaw while excluding roll and pitch from the vehicle state. This approximation is motivated by the operating characteristics of survey-class UUVs during steady-state seabed mapping missions, where roll and pitch angles typically remain small in the absence of aggressive maneuvers. Fig. 2 illustrates the geometric relationship between the vehicle and landmark states under the reduced 4-DOF representation, where the azimuth measurement is mainly associated with the horizontal relative geometry, while the slant-range measurement reflects the full 3D separation. The system dynamics are modeled as a nonlinear stochastic process, expressed as

$$\dot{\mathbf{x}}_v \begin{bmatrix}\dot{x}_v \\ \dot{y}_v \\ z_v \\ \dot{\psi}_v\end{bmatrix} = \begin{bmatrix}v_x \cos\psi_v - v_y \sin\psi_v \\ v_x \sin\psi_v + v_y \cos\psi_v \\ v_z \\ \omega_\psi\end{bmatrix} + \mathbf{w} \tag{7}$$

where $v_x, v_y,$ and v_z denote, the linear velocities along the x-, y-, and z- axis repectively, and ω_ψ indicates the yaw rate. These quantities form the control input vector $u = [v_x \ v_y \ v_z \ \omega_\psi]^\top$. The linear velocities can be measured using a Doppler velocity log (DVL), while the yaw rate is obtained from an inertial measurement unit (IMU). The process noise vector $\mathbf{w}$ is assumed to follow a zero-mean Gaussian distribution.

3. Simulation Study

3.1 Simulation Setup and Scenario

To evaluate the proposed formulation in SSS SLAM, simulation experiments are conducted using Stonefish [7], a physics-based underwater simulation environment that supports multiple sensing modalities, including SSS. The simulated UUV is equipped with an IMU, a DVL, and an SSS. The IMU and DVL operate at 50 Hz, while the SSS provides ping measurements at 10 Hz with a maximum sensing range of 100 m. The survey area covers a $300\ \mathrm{m} \times 300\ \mathrm{m}$ region, where the seabed terrain exhibits height variations of approximately 3 m, which are negligible relative to the overall spatial scale. As illustrated in Fig. 3, a set of fixed objects is placed at predefined locations in the global frame, slightly protruding from an otherwise homogeneous seabed. These objects generate distinguishable acoustic backscatter responses and are treated as landmarks for SLAM. The simulated UUV follows three predefined lawnmower survey trajectories designed to repeatedly observe the same landmarks from different viewpoints. Examples of the sensing configuration and the resulting SSS imagery are shown in Fig. 4. In the simulated survey trajectories, the vehicle roll and pitch remained within approximately $-0.16°$ to $0.02°$ and $-0.29°$ to $0.33°$, respectively, consistent with the small-angle approximation ($\phi_v \approx 0$ and $\theta_v \approx 0$).

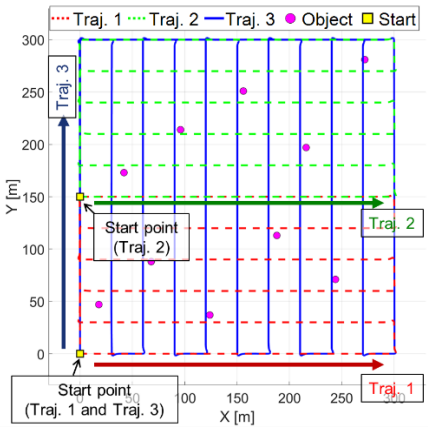
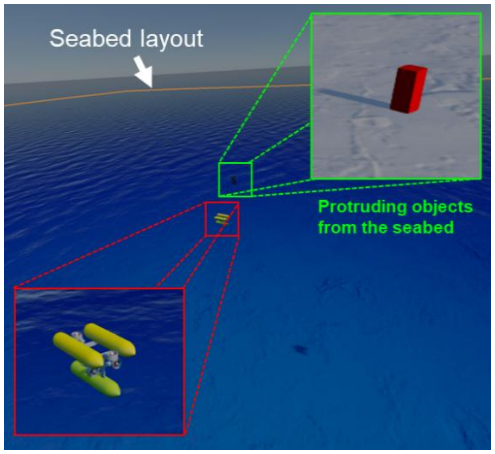
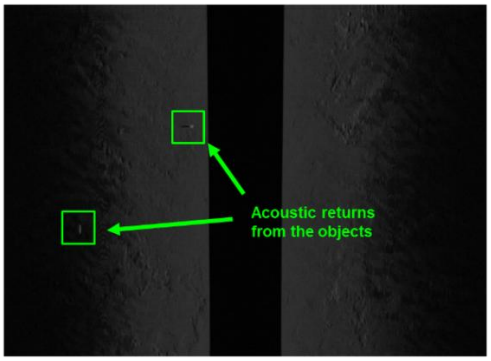


Fig. 3 Simulation scenario showing the placement of fixed landmarks and the three predefined lawnmower survey trajectories used for evaluation.



(a)



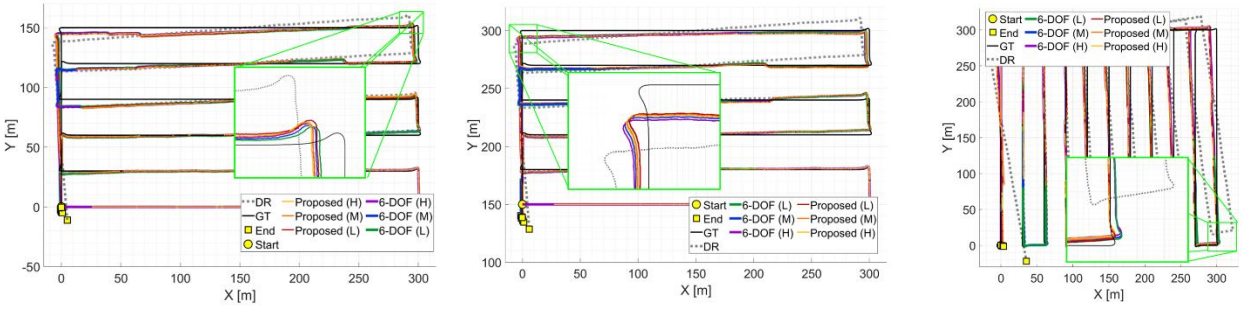
(b)

Fig. 4 Examples from `Traj. 1`. (a) Snapshot of the simulated survey environment showing objects protruding from an otherwise homogeneous seabed. (b) Corresponding SSS image generated from ping measurements, where the protruding objects produce distinguishable acoustic backscatter responses appearing as localized intensity regions.

For performance evaluation, the proposed formulation is compared with a conventional 6-DOF state representation under identical simulation conditions. Both formulations are implemented within the same EKF-based SLAM framework and use the same measurements, data association results, and the same process and measurement noise settings for the common state components, ensuring that any performance difference can be attributed solely to the state representation. To investigate how landmark elevation uncertainty influences state estimation, the azimuth uncertainty σ_α is varied at three levels (1°, 2°, and 4°, denoted as L, M, H, respectively). This analysis particularly focuses on the azimuth component of the SSS observation, since the approximated azimuth Jacobian in Eq. (5) contains a nonzero pitch-related term associated with the landmark elevation, whereas the approximated slant-range Jacobian contains zero entries for the attitude states under the assumed sensing geometry. This implies that the slant-range measurement does not directly contribute to attitude-state updates, while the azimuth measurement can propagate landmark elevation uncertainty into the pitch-state estimation process. Newly observed landmark states are initialized using the corresponding SSS measurement together with the vehicle's estimated 3D position and nominal altitude, assuming that the landmark is located near the seabed. Since SSS measurements do not directly constrain landmark elevation, a relatively large initial covariance is assigned to the vertical component of the landmark state to reflect this elevation degeneracy and preserve conservative uncertainty estimates in the vertical direction during filtering. Note that this landmark elevation uncertainty is associated with the landmark initialization and is distinct from the azimuth measurement uncertainty σ_α varied in the simulation study. Estimation accuracy is evaluated using the ATE (absolute trajectory error) of both the vehicle and landmark states. Specifically, the evaluation separately considers the horizontal vehicle position (x_v, y_v), vehicle depth (i.e., vertical position) z_v, horizontal landmark position (x_ℓ, y_ℓ), and landmark elevation z_ℓ. All metrics are computed over the three survey trajectories to assess the accuracy of each formulation under different observation geometries.

3.2 Results

Fig. 5 shows the qualitative horizontal trajectory estimation results for the three simulated survey trajectories. Both the conventional 6-DOF and proposed 4-DOF formulations successfully recover the overall lawnmower trajectory patterns under different azimuth uncertainty settings. Quantitative results in Table 1 further show that the horizontal localization performance of the two formulations remains comparable in terms of both the vehicle



`Traj. 1` `Traj. 2` `Traj. 3`

Fig. 5 Qualitative comparison of horizontal trajectory estimation results across three simulated survey trajectories. Both the conventional 6-DOF and proposed 4-DOF formulations successfully recover the overall lawnmower trajectory patterns under different azimuth uncertainty settings, yielding comparable horizontal localization performance despite the different state representations and observation models..

horizontal position (x_v, y_v) and landmark horizontal position (x_ℓ, y_ℓ). Although certain trajectories and uncertainty settings show slightly lower horizontal errors for either formulation, the overall difference remains limited. This behavior can be attributed to the different observation models used in the two formulations, where the conventional 6-DOF formulation directly utilizes the full 3D geometry of the slant-range and azimuth observation model, while the proposed formulation adopts a reduced state representation based on the dominant sensitivity components of the sensing geometry.

In contrast, the two formulations exhibit substantially different behavior in the vertical states. As summarized in Table 1, the conventional 6-DOF formulation consistently produces significantly larger errors in both the vehicle depth z_v and landmark elevation z_ℓ compared with the proposed formulation across all trajectories. This degradation mechanism is further illustrated in Fig. 6, which shows the estimated pitch trajectories of the DR and conventional 6-DOF formulation. Note that the proposed 4-DOF formulation is omitted because pitch is not included in its state to be estimated. As the azimuth uncertainty decreases, the azimuth measurement update is assigned higher confidence, leading to larger pitch fluctuations in the conventional 6-DOF formulation due to the elevation-dependent sensitivity of the measurement Jacobian discussed in Section 2. Fig.7 illustrates how these pitch fluctuations affect the vehicle depth estimation. In the conventional 6-DOF formulation, the vehicle depth estimates exhibit progressively larger drift and unstable behavior under lower azimuth uncertainty settings, despite the higher confidence assigned to the measurements. This behavior indicates that the underlying elevation degeneracy remains structurally unresolved in the conventional formulation, and that

Table 1 Estimation performance comparison across three survey trajectories under different vehicle state representations and azimuth uncertainty settings. The metrics report the ATE of the vehicle horizontal position $(\boldsymbol{x_v, y_v})$, vehicle depth $\boldsymbol{z_v}$, landmark horizontal position $(\boldsymbol{x_\ell, y_\ell})$, and landmark elevation $\boldsymbol{z_\ell}$. L, M, and H denote low, medium, and high azimuth uncertainty settings corresponding to $\boldsymbol{\sigma_\alpha = 1}$°, $\boldsymbol{2}$°, and $\boldsymbol{4}$°, respectively. Note that minor numerical differences appearing beyond the second decimal place are considered negligible.

Method	Traj. 1				Traj. 2				Traj. 3			
	ATE [m]				ATE [m]				ATE [m]			
	(x_v, y_v)	z_v	(x_ℓ, y_ℓ)	z_ℓ	(x_v, y_v)	z_v	(x_ℓ, y_ℓ)	z_ℓ	(x_v, y_v)	z_v	(x_ℓ, y_ℓ)	z_ℓ
DR	6.06	1.77	-	-	5.99	1.79	-	-	15.74	3.28	-	-
6-DOF (L)	3.32	4.47	2.59	4.62	3.24	5.01	1.85	3.09	4.18	10.28	3.27	7.30
6-DOF (M)	3.61	3.86	3.34	4.05	3.36	3.74	2.03	2.65	4.36	10.13	3.32	7.38
6-DOF (H)	3.80	3.68	3.94	3.87	3.47	3.52	2.11	2.49	4.63	8.31	3.41	6.18
Proposed (L)	3.90	0.73	2.56	1.96	3.16	0.73	1.80	1.41	4.61	1.08	2.92	1.04
Proposed (M)	3.97	0.73	3.53	1.91	3.21	0.73	1.97	1.45	4.90	1.08	2.94	1.03
Proposed (H)	4.06	0.73	4.06	1.98	3.25	0.73	2.05	1.50	5.10	1.08	2.98	1.05

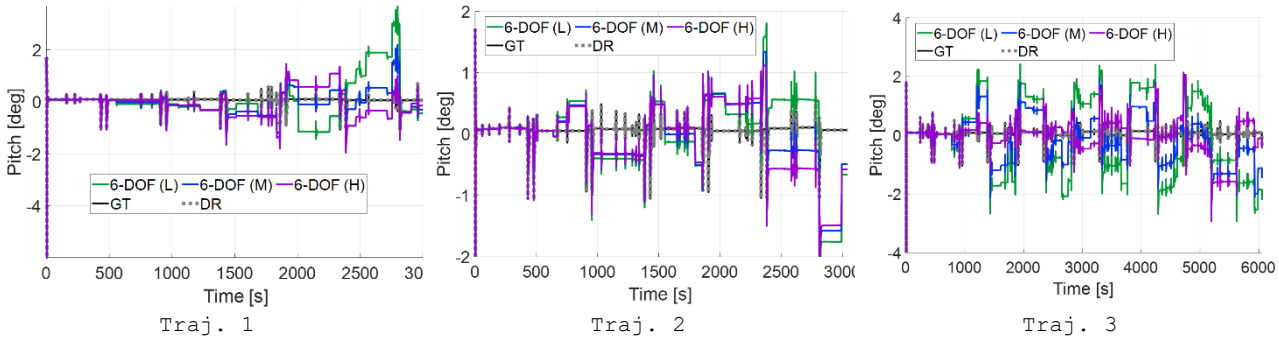


Fig. 6 Pitch estimation results of the DR and conventional 6-DOF formulation across three simulated survey trajectories under different azimuth uncertainty settings. Note that the proposed 4-DOF formulation is omitted because pitch is not included in its estimation state.

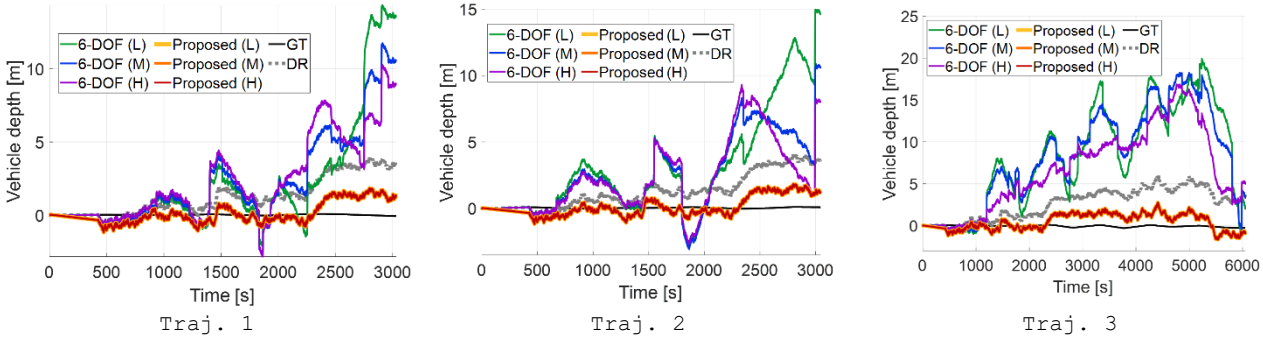


Fig. 7 Time histories of the vehicle depth across three simulated survey trajectories. In the conventional 6-DOF formulation, fluctuations in the estimated landmark elevation propagate into the pitch-state update, inducing unstable coupling between the vehicle depth and pitch states and ultimately increasing estimation errors. By contrast, the proposed 4-DOF formulation maintains stable depth estimation behavior across different azimuth uncertainty settings.

assigning excessive confidence to the azimuth measurement update amplifies the resulting estimation errors. In contrast, the proposed 4-DOF formulation maintains nearly invariant z_v-ATE values across different azimuth uncertainty settings because roll and pitch are excluded from the estimation state.

The same trend is also observed in the reconstructed landmark map. As shown in Table 1, the conventional 6-DOF formulation produces substantially larger landmark elevation errors (i.e., in z_ℓ) than the proposed formulation across all trajectories and uncertainty settings. This is further visualized in Fig. 8, which compares the landmark elevation ATE for `Traj. 3`. In the conventional 6-DOF formulation, landmarks detected later in the mission exhibit progressively larger vertical errors due to the accumulated drift in the vehicle vertical-state estimation. By contrast, the proposed formulation maintains consistently low landmark elevation errors across all landmarks and azimuth uncertainty settings, demonstrating significantly more stable vertical-state estimation behavior under elevation degeneracy.

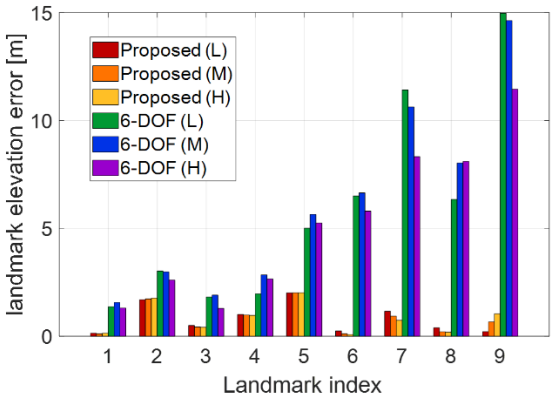


Fig. 8 Comparison of landmark elevation ATE for `Traj. 3` under different azimuth uncertainty settings. The proposed formulation maintains uniformly low landmark elevation errors across all landmarks and uncertainty settings, whereas the conventional 6-DOF formulation exhibits progressively larger errors for landmarks observed later in the mission, reflecting the accumulated growth of vertical-state estimation drift.

4. Conclusion

This paper investigated the effect of landmark elevation uncertainty on state estimation in landmark-based SSS SLAM under elevation degeneracy. Through Jacobian analysis of the slant-range and azimuth observation model, the relationship between the native sensing geometry of SSS measurements and the resulting state-estimation behavior was examined. In particular, the analysis showed that uncertainty in landmark elevation can propagate into the vehicle pitch-state update in conventional full 6-DOF

formulations, potentially degrading the accuracy of vertical-state estimation. Based on this analysis, a reduced 4-DOF vehicle state representation excluding roll and pitch was considered according to the state sensitivities imposed by the sensing geometry. The effectiveness and performance of landmark-based SSS SLAM with the proposed state representation was demonstrated through comparative physics-based underwater simulation experiments against a conventional 6-DOF formulation under different measurement uncertainty conditions.

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